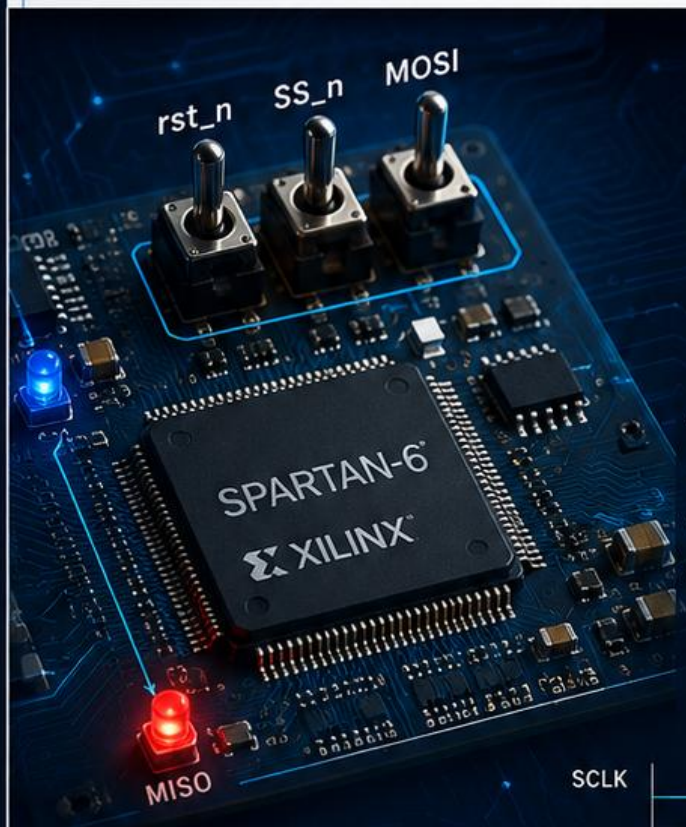


DIGITAL ELECTRONICS PROJECT

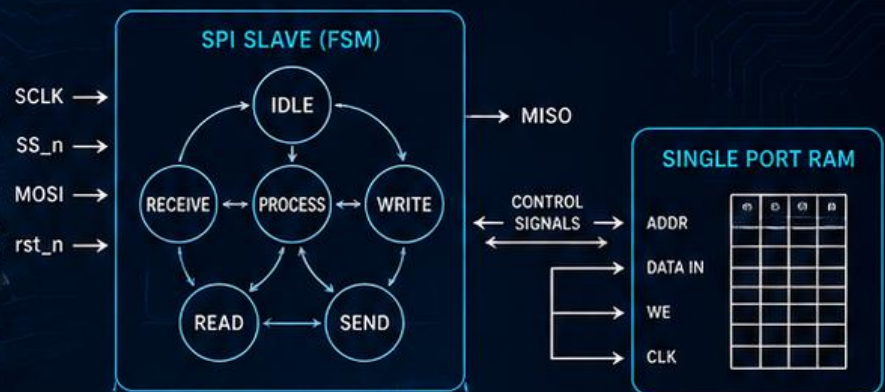
SPI SLAVE WITH SINGLE PORT RAM

DESIGN • VERIFICATION • IMPLEMENTATION



SPI SLAVE WITH SINGLE PORT RAM

VERILOG • FPGA • FSM DESIGN



SPI COMMUNICATION



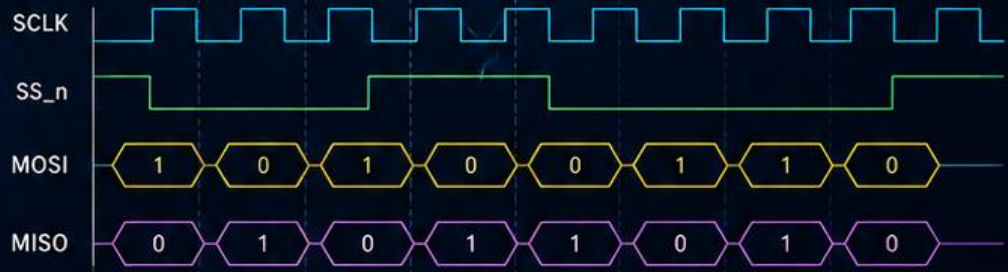
FSM CONTROL



RAM READ / WRITE



FPGA IMPLEMENTATION



Prepared by:
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Supervised by:
Eng. Kareem Waseem

SPI code :

```
module SPI (
    MOSI , SS_n , clk , rst_n , tx_data , tx_valid , MISO , rx_valid , rx_data
);
input MOSI , SS_n , clk , rst_n , tx_valid ;
input [7:0] tx_data ;
output [9:0] rx_data;
output reg MISO , rx_valid ;

// parameters
parameter IDLE      = 3'b000 ;
parameter CHK_CMD   = 3'b001 ;
parameter WRITE     = 3'b010 ;
parameter READ_ADD  = 3'b011 ;
parameter READ_DATA = 3'b100 ;

reg address_entered ; // make choice between read_data , read_add
reg [3:0] counter ;
reg [9:0] data_reg ;

(* fsm_encoding = "one_hot" *)
reg [2:0] cs, ns;

//next state
always @(*) begin
    case (cs)

        IDLE: ns = (SS_n == 0) ? CHK_CMD : IDLE;

        CHK_CMD: begin
            if (SS_n == 1)
                ns = IDLE;
            else begin
                if (MOSI == 1) begin
                    if (address_entered == 1)
                        ns = READ_DATA;
                    else
                        ns = READ_ADD;
                end
                else
                    ns = WRITE;
            end
        end

        WRITE:
            ns = (SS_n == 1) ? IDLE : WRITE;

        READ_ADD:
            ns = (SS_n == 1) ? IDLE : READ_ADD;
```

```

        READ_DATA:
            ns = (SS_n == 1) ? IDLE : READ_DATA;

        default: ns = IDLE;

    endcase
end

// state memory
always @(posedge clk) begin
    if(~rst_n)
        cs <= IDLE ;
    else
        cs <= ns    ;
end

// Output
always @(posedge clk) begin

    if (~rst_n) begin
        MISO <= 0;

        rx_valid <= 0;
        counter <= 0;
        data_reg <= 0;
        address_entered <= 0;
    end

    else begin
        case(cs)

            IDLE: begin
                rx_valid <= 0;
            end

            CHK_CMD: begin
                counter <= 10;
            end

            WRITE: begin

                if(counter > 0) begin
                    data_reg[counter-1] <= MOSI;
                    counter <= counter - 1;
                end

                else begin

```

```

        rx_valid <= 1;

    end

end

READ_ADD: begin

    if(counter > 0) begin
        data_reg[counter-1] <= MOSI;
        counter <= counter - 1;
    end

    else begin
        rx_valid <= 1;
        address_entered <= 1;

    end

end

READ_DATA: begin

    if(tx_valid) begin

        rx_valid <= 0;

        if(counter > 0) begin
            MISO <= tx_data[counter-1];
            counter <= counter - 1;
        end

        else begin

            address_entered <= 0;
        end

    end

    else begin

        if(counter > 0) begin
            data_reg[counter-1] <= MOSI;
            counter <= counter - 1;
        end

        else begin
            rx_valid <= 1;

            counter <= 9;
        end
    end
end

```

```

        end

    end

    default: begin
        rx_valid <= 0;
    end

endcase

end

end
assign rx_data = data_reg;

endmodule

```

RAM code :

```

module RAM (
    din ,rst_n , clk , rx_valid , dout , tx_valid
);
parameter MEM_DEPTH = 256 ;
parameter ADDR_SIZE = 8 ;

input [9:0] din ;
input clk , rst_n , rx_valid ;

output reg tx_valid ;
output reg [ADDR_SIZE-1 : 0 ] dout ;

reg [ADDR_SIZE-1 : 0] wr_addr , rd_addr ;
reg [ADDR_SIZE-1 : 0] mem [MEM_DEPTH - 1 : 0] ;

always @(posedge clk) begin
    if (~rst_n)begin
        dout      <= 0 ;
        tx_valid  <= 0 ;
        wr_addr   <= 0 ;
        rd_addr   <= 0 ;
    end
    else begin
        if (rx_valid) begin
            case (din[9:8])

                2'b00: begin
                    wr_addr <= din [ADDR_SIZE -1 : 0] ;
                    tx_valid<= 0 ;
                end
                2'b01: begin

```

```

        mem [wr_addr] <= din [ADDR_SIZE -1 : 0] ;
        tx_valid<= 0 ;
    end
    2'b10: begin
        rd_addr <= din [ADDR_SIZE -1 : 0] ;
        tx_valid<= 0 ;
    end
    2'b11: begin
        dout      <= mem [rd_addr] ;
        tx_valid<= 1 ;
    end

    endcase
end
end
endmodule

```

TOP wrapper code :

```

module SPI_RAM_TOP (
    input MOSI,
    input SS_n,
    input clk,
    input rst_n,
    output MISO
);

// internal wires
wire [9:0] rx_data;
wire rx_valid;

wire [7:0] tx_data;
wire tx_valid;

// SPI instantiation
SPI SPI_inst (

    .MOSI(MOSI),
    .SS_n(SS_n),
    .clk(clk),
    .rst_n(rst_n),

    .tx_data(tx_data),
    .tx_valid(tx_valid),

    .MISO(MISO),

    .rx_valid(rx_valid),
    .rx_data(rx_data)
);

```

```
// RAM instantiation
RAM RAM_inst (

    .din(rx_data),
    .clk(clk),
    .rst_n(rst_n),

    .rx_valid(rx_valid),

    .dout(tx_data),
    .tx_valid(tx_valid)

);

endmodule
```

testbench code :

```
module SPI_tb ();
reg MOSI, SS_n, clk, rst_n;
wire MISO;

reg [9:0] MOSI_tb;

// DUT
SPI_RAM_TOP DUT (
    .MOSI(MOSI),
    .SS_n(SS_n),
    .clk(clk),
    .rst_n(rst_n),
    .MISO(MISO)
);

initial begin
    clk = 0;
    forever #1 clk = ~clk;
end

integer i;

initial begin
    $readmemh("mem.dat", DUT.RAM_inst.mem);

    rst_n = 0;
    MOSI = 0;
    SS_n = 1;
    MOSI_tb = 0;

    //Check Reset function
    @(negedge clk);
    if(MISO != 0) begin
```

```

        $display("Error");
        $stop;
    end

    //Check Write address function
    rst_n = 1;
    SS_n = 0;
    @(negedge clk);
    MOSI = 0;

    @(negedge clk);
    MOSI_tb = 10'b00_00100101 ; // 00 8'd37

    for (i = 0 ; i < 10 ; i = i + 1 ) begin
        MOSI = MOSI_tb[9-i];
        @(negedge clk);
    end

    @(negedge clk);

    //Check Write Data function
    SS_n = 1;
    @(negedge clk);
    SS_n = 0;
    @(negedge clk);
    MOSI = 0;
    @(negedge clk);

    MOSI_tb = 10'b01_01100100 ; // mem[37] = 100

    for ( i = 0 ; i < 10 ; i = i + 1 ) begin
        MOSI = MOSI_tb[9-i];
        @(negedge clk);
    end

    @(negedge clk);
    @(negedge clk);

    //Check read address in memory
    SS_n = 1;
    @(negedge clk);
    SS_n = 0;
    @(negedge clk);
    MOSI = 1;
    @(negedge clk);

    MOSI_tb = 10'b10_00100101 ; // addr_rd = 37

    for ( i = 0 ; i < 10 ; i = i + 1 ) begin
        MOSI = MOSI_tb[9-i];
        @(negedge clk);
    end
end

```



```
//Check read data from memory
SS_n = 1;
@(negedge clk);
SS_n = 0;
@(negedge clk);
MOSI = 1;
@(negedge clk);

MOSI_tb = 10'b11_00100101 ;

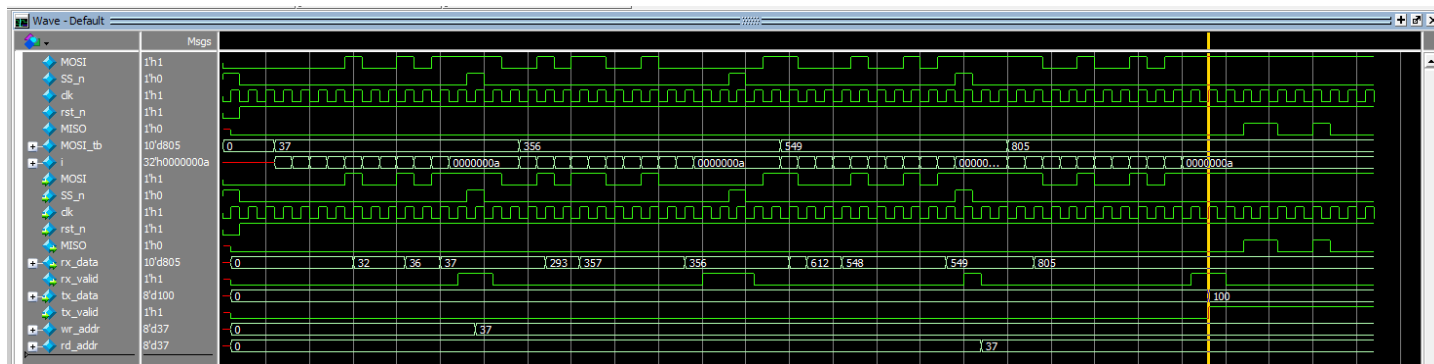
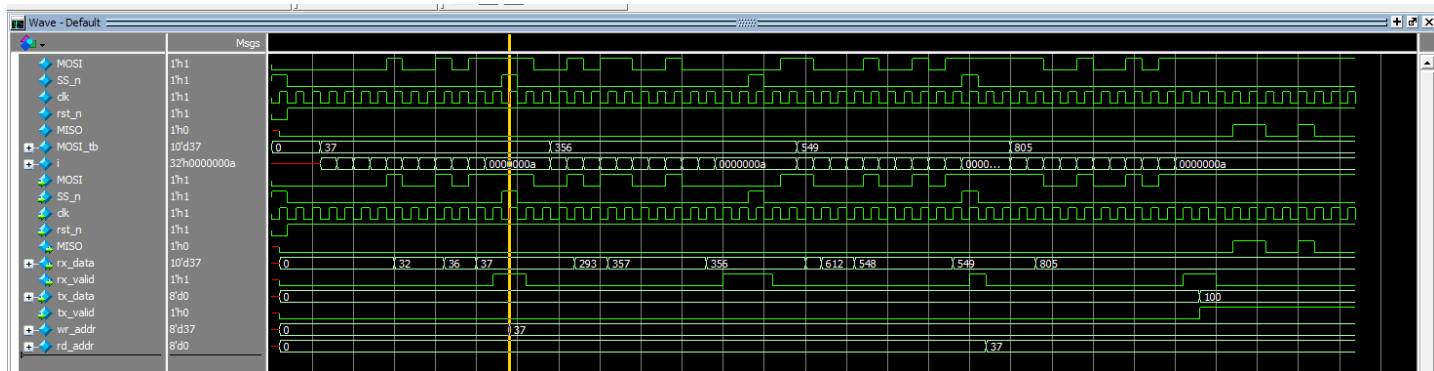
for ( i = 0 ; i < 10 ; i = i + 1 ) begin
    MOSI = MOSI_tb[9-i];
    @(negedge clk);
end

repeat(11) @(negedge clk);

$stop;
end

endmodule
```

QuestaSim:



QuestaLint:

Questa Lint 2021.1 (J:\Desktop\QuestaLint_proj\SPL_proj\lint.db)

File Edit View Lint Summary Window Help

Design

Search: Type Sea... Exact

Instance: SPI_RAM_TOP(2) Module: SPI_RAM_TOP

Flow Navigator Design

```
1 module SPI_RAM_TOP (  
2     input MOSI,  
3     input SS_n,  
4     input clk,  
5     input rst_n,  
6     output MISO  
7 );  
8
```

Lint Summary

(Type Search Text (Press Enter))

Name	Count
Open(unspected, pending, bug)	2 (3)
Info	2 (3)

Lint Checks

Filter: Type here

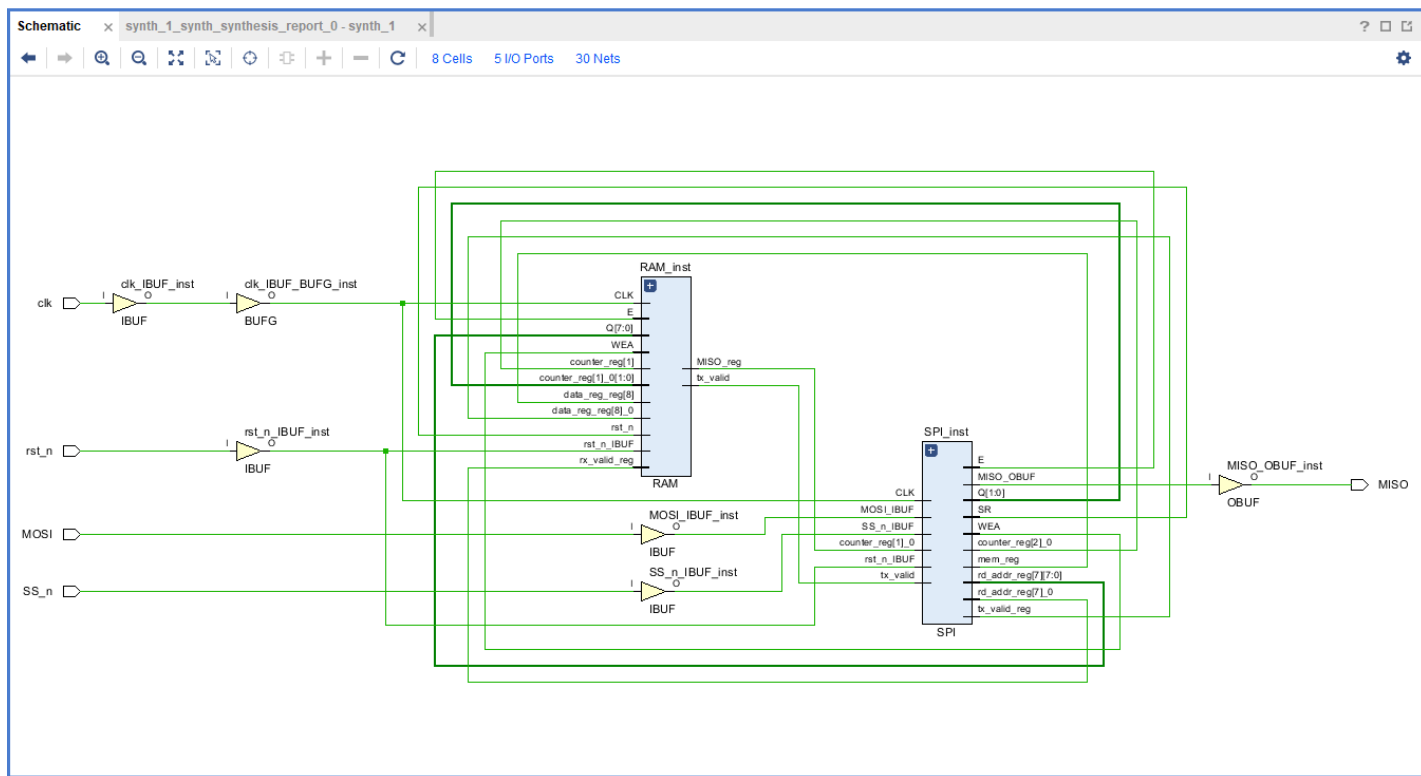
Severity Status Check Alias Message Module Category State Owner STARC Reference

Warning	Warning	multi_ports_in_single_line		Multiple ports are declared in one line. Module RAM, File C:/Users/m...	RAM	Rtl Design Style	open	unassigned	3.5.6.3	
Warning	Warning	multi_ports_in_single_line		Multiple ports are declared in one line. Module SPI, File C:/Users/ma...	SPI	Rtl Design Style	open	unassigned	3.5.6.3	

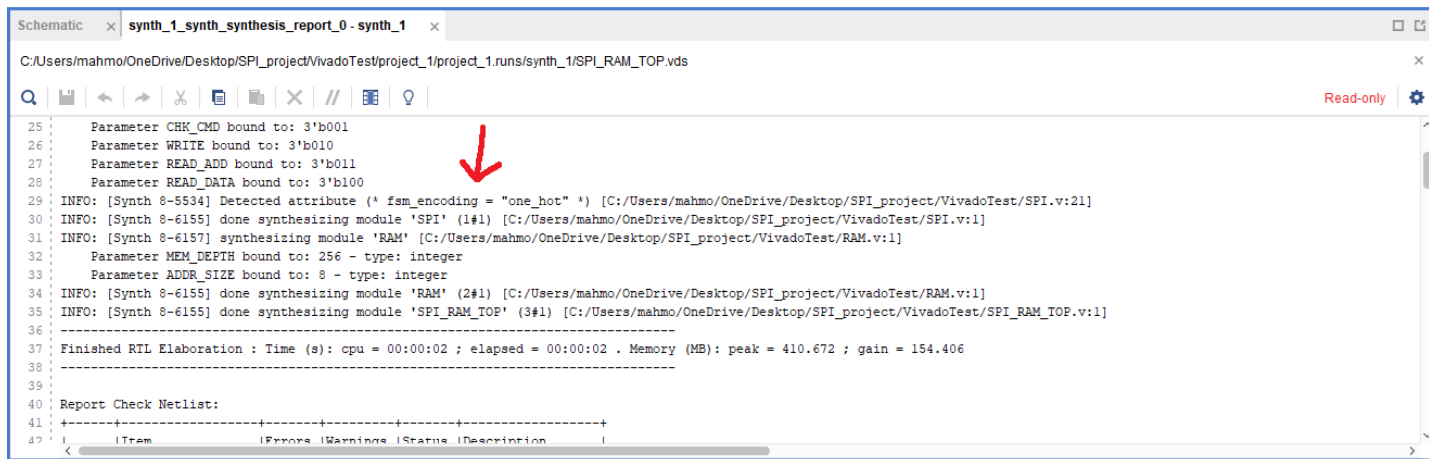
Vivado:-

One_hot:

Schematic:



Synthesis Report:



Timing Report:

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 6.261 ns	Worst Hold Slack (WHS): 0.149 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 87	Total Number of Endpoints: 87	Total Number of Endpoints: 42

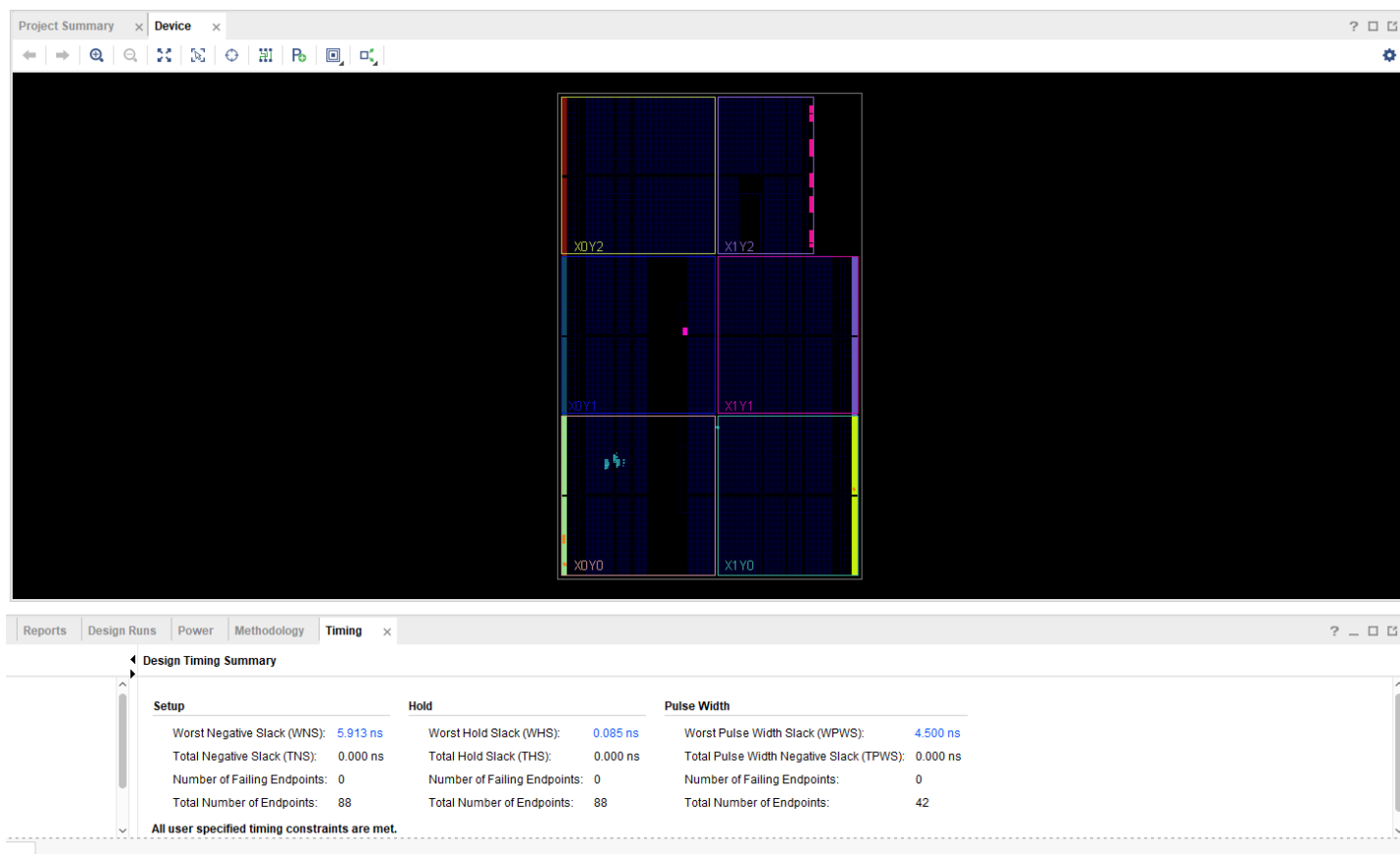
All user specified timing constraints are met.

Critical Path:

The screenshot shows the Vivado Synthesis Report for the project 'synth_1_synth_synthesis_report_0 - synth_1'. The report details the critical path analysis. The critical path is identified as Path 1, with a worst negative slack of 6.261 ns. The report also shows the synthesis of the 'RAM' module and the 'SPI' module. The final RTL Elaboration time is 00:00:02, and the peak memory usage is 410.672 MB.

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception
Path 1	6.261	2	3	1	RAM_inst/mem_reg/CLKBWRCLK	SPI_inst/MISO_reg/D	3.623	2.823	0.800	10.0	sys_clk_pin	sys_clk_pin	
Path 2	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[0]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 3	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[1]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 4	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[2]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 5	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[3]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 6	6.964	1	2	6	SPI_inst/rx_valid_reg/C	RAM_inst/mem_reg/WEA[0]	2.324	0.751	1.573	10.0	sys_clk_pin	sys_clk_pin	
Path 7	6.964	1	2	6	SPI_inst/rx_valid_reg/C	RAM_inst/mem_reg/WEA[1]	2.324	0.751	1.573	10.0	sys_clk_pin	sys_clk_pin	
Path 8	6.987	2	3	13	SPI_inst/FSM_onehot_cs_reg[2]/C	SPI_inst/data_reg[0]/CE	2.631	0.875	1.756	10.0	sys_clk_pin	sys_clk_pin	
Path 9	6.987	2	3	13	SPI_inst/FSM_onehot_cs_reg[2]/C	SPI_inst/data_reg[1]/CE	2.631	0.875	1.756	10.0	sys_clk_pin	sys_clk_pin	
Path 10	6.987	2	3	13	SPI_inst/FSM_onehot_cs_reg[2]/C	SPI_inst/data_reg[2]/CE	2.631	0.875	1.756	10.0	sys_clk_pin	sys_clk_pin	

Implementation:

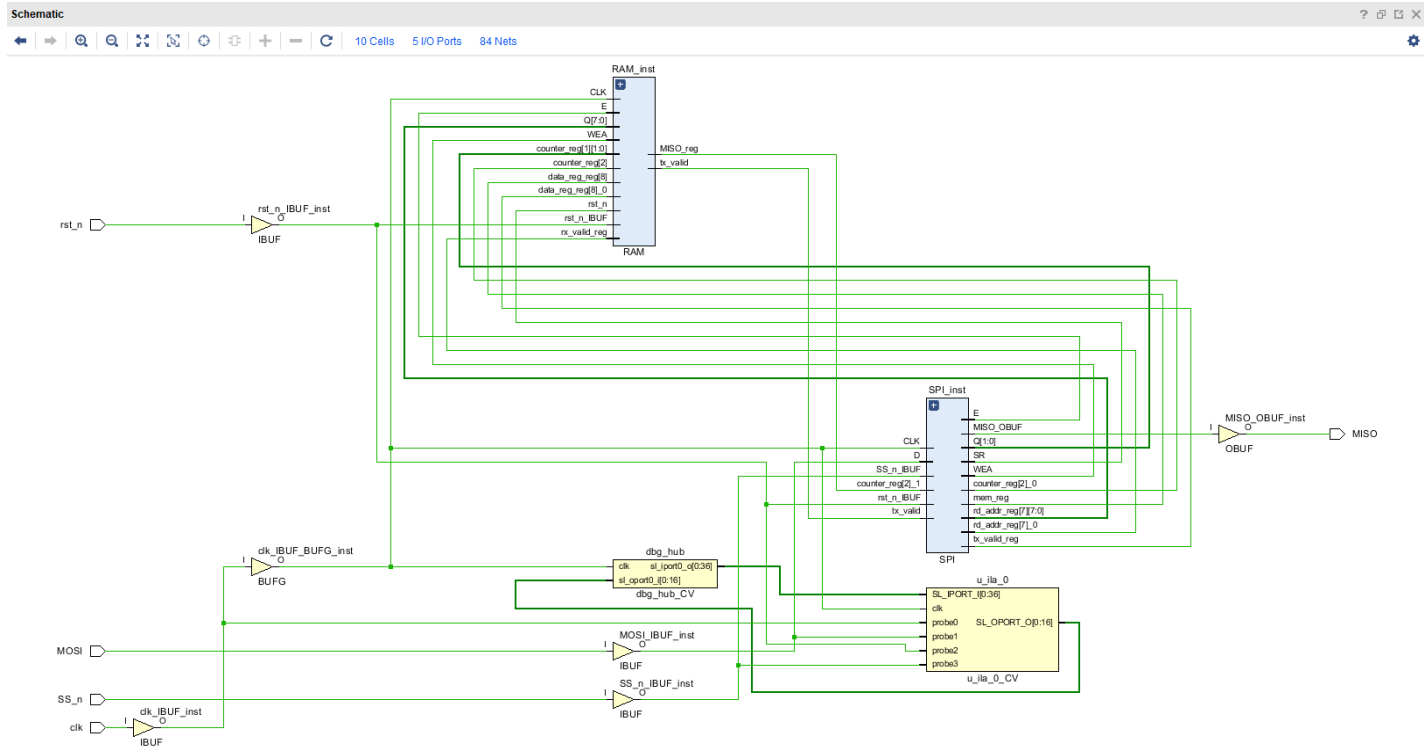


Utilization Report:

Design Runs		Power	Methodology	Timing	Utilization					
Hierarchy										
Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (8150)	LUT as Logic (20800)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)	BUFGCTRL (32)	
▼ N SPI_RAM_TOP	38	39	1	21	38	12	0.5	5	1	
RAM_inst (RAM)	3	17	1	6	3	0	0.5	0	0	
SPI_inst (SPI)	35	22	0	20	35	11	0	0	0	

gray:

Schematic:



Synthesis Report:

```
Schematic x synth_1_synth_synthesis_report_0 - synth_1 x
C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/project_1/runs/synth_1/SPI_RAM_TOP.vds

27 Parameter READ_ADD bound to: 3'b011
28 Parameter READ_DATA bound to: 3'b100
29 INFO: [Synth 8-5534] Detected attribute (* fsm_encoding = "gray" *) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI.v:21]
30 INFO: [Synth 8-6155] done synthesizing module 'SPI' (1#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI.v:1]
31 INFO: [Synth 8-6157] synthesizing module 'RAM' [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/RAM.v:1]
32 Parameter MEM_DEPTH bound to: 256 - type: integer
33 Parameter ADDR_SIZE bound to: 8 - type: integer
34 INFO: [Synth 8-6155] done synthesizing module 'RAM' (2#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/RAM.v:1]
35 INFO: [Synth 8-6155] done synthesizing module 'SPI_RAM_TOP' (3#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI_RAM_TOP.v:1]
36
37 Finished RTL Elaboration : Time (s): cpu = 00:00:02 ; elapsed = 00:00:02 . Memory (MB): peak = 410.172 ; gain = 153.047
38
39
40 Report Check Netlist:
41
```

Timing Report:

Runs

Timing

×

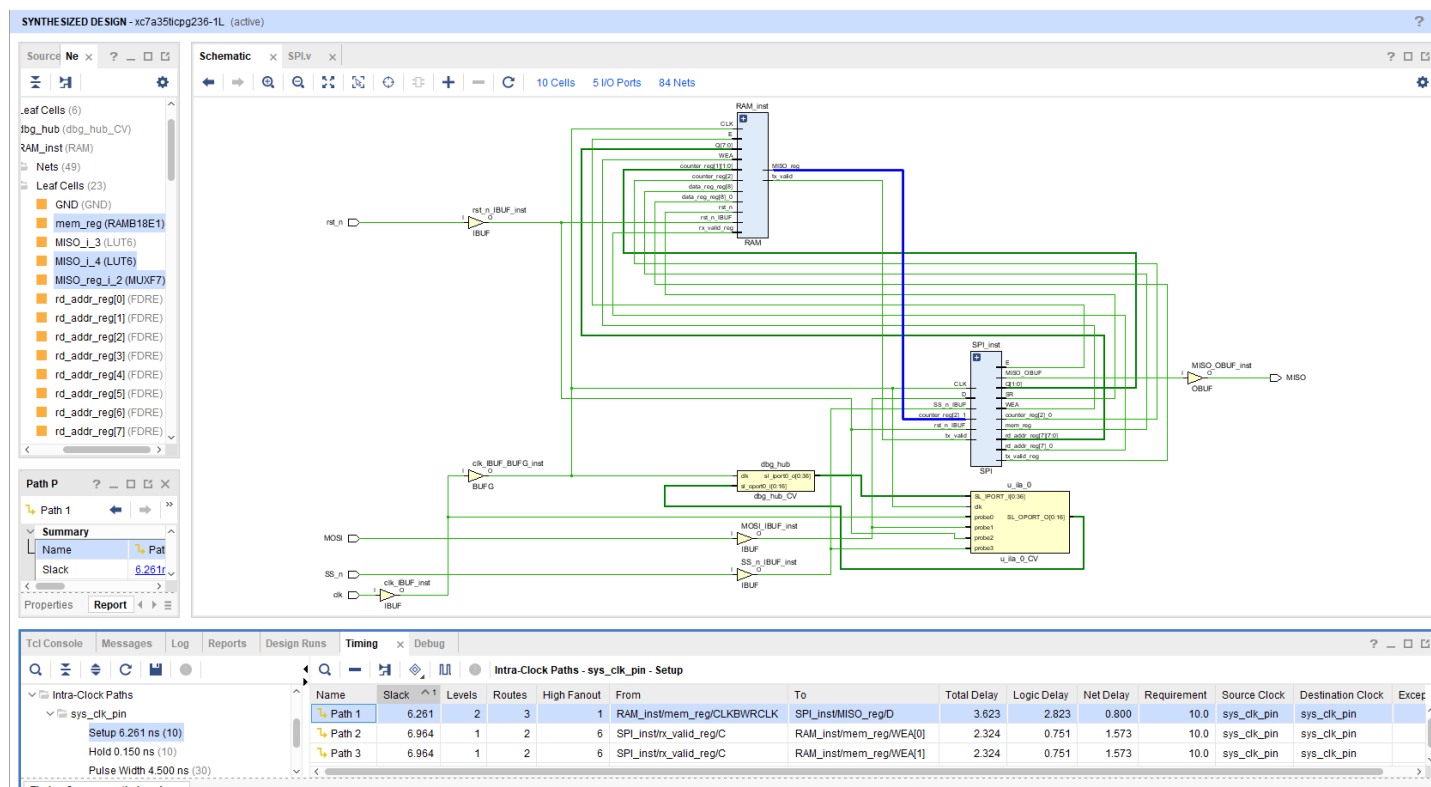
Debug

Design Timing Summary

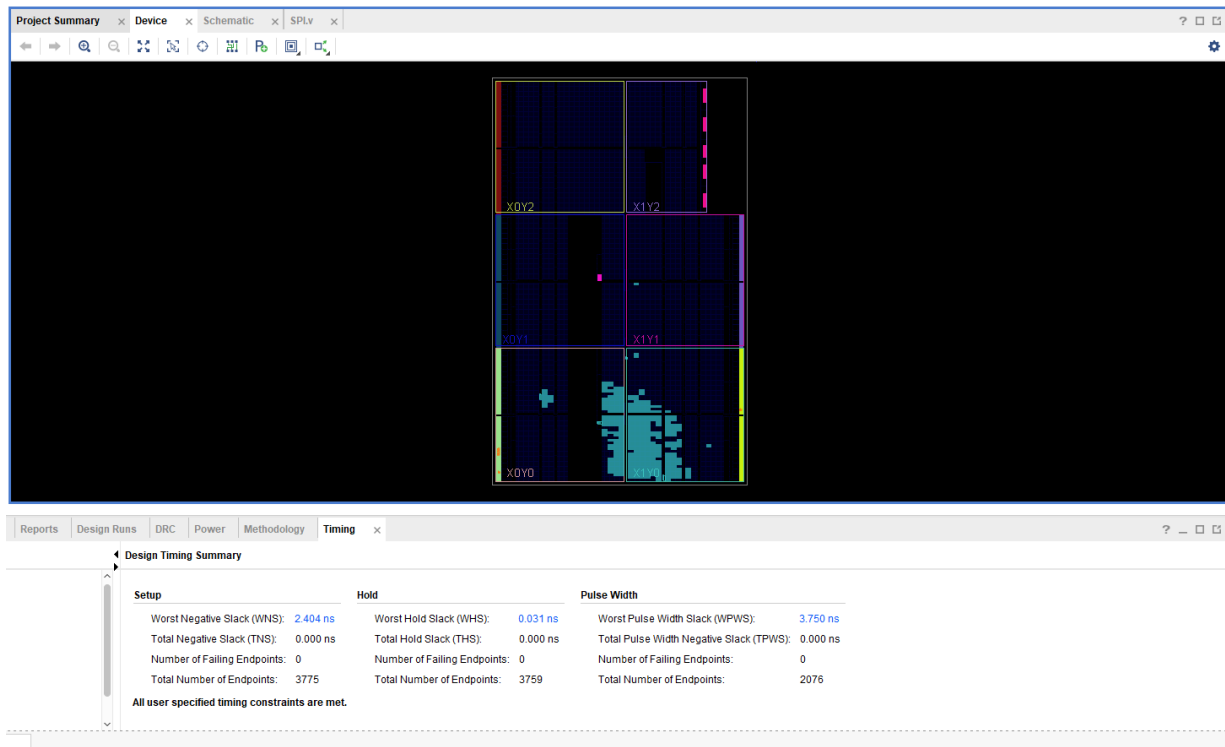
Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 6.261 ns	Worst Hold Slack (WHS): 0.150 ns	Worst Pulse Width Slack (WPWS): 4.500 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 85	Total Number of Endpoints: 85	Total Number of Endpoints: 40

All user specified timing constraints are met.

Critical Path:



Implementation:



Utilization Report:

Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (815 0)	LUT as Logic (20800)	LUT as Memory (9600)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)	BUFCTRL (32)	BSCANES2 (4)
SPI_RAM_TOP	1221	1899	9	607	1118	103	712	1	5	2	1
dbg_hub (dbg_hub)	475	727	0	235	451	24	313	0	0	1	1
RAM_inst (RAM)	2	17	1	4	2	0	0	0.5	0	0	0
SPI_inst (SPI)	30	20	0	17	30	0	9	0	0	0	0
u_ila_0 (u_ila_0)	714	1135	8	357	635	79	388	0.5	0	0	0

Messages tab with no critical warnings or errors:

Tcl Console Messages x Log Reports Design Runs Timing Debug

Warning (5) Info (481) Status (988) Show All

- Synthesis (1 warning)
 - [Constraints 18-5210] No constraint will be written out.
- Implementation (2 warnings)
 - Route Design (1 warning)
 - [Timing 38-436] There are set_bus_skew constraint(s) in this design. Please run report_bus_skew to ensure that bus skew requirements are met.
 - Write Bitstream (1 warning)
 - DRC (1 warning)
 - Implementation (1 warning)
 - Routing (1 warning)
 - Chip Level (1 warning)
 - [DRC RTSTAT-10] No routable loads: 25 net(s) have no routable loads. The problem bus(es) and/or net(s) are
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD6_RD0U_RD_FIFOSUBCORE_FIF0_xsdm_v3_0_0_rdfifo_instInst_fifo_genlgconvffo_rdfgr_rfgnfv_or_sync_fifo_glo_rdfgrt_grt_int_rfwf/empty_fwft_i,
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD7_CTLctl_reg2_0, dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD7_CTLctl_reg_en_2[1],
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD1ctl_reg_en_2[1], dbg_hubInstBSCANID_u_xsdm_idSWITCH_N_EXT_BSCAN_bscan_switchm_bscan_capture[0],
dbg_hubInstBSCANID_u_xsdm_idSWITCH_N_EXT_BSCAN_bscan_switchm_bscan_dro[0], dbg_hubInstBSCANID_u_xsdm_idSWITCH_N_EXT_BSCAN_bscan_switchm_bscan_runtest[0],
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD6_RD0U_RD_FIFOSUBCORE_FIF0_xsdm_v3_0_0_rdfifo_instInst_fifo_genlgconvffo_rdfgr_rfgnfv_or_sync_fifo_glo_wrigwht_whyoverflow,
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD6_RD0U_RD_FIFOSUBCORE_FIF0_xsdm_v3_0_0_rdfifo_instInst_fifo_genlgconvffo_rdfgr_rfgnfv_or_sync_fifo_glo_wrigwht_whyoverflow,
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD6_WRUU_WR_FIFOSUBCORE_FIF0_xsdm_v3_0_0_wrtffo_instInst_fifo_genlgconvffo_rdfgr_rfgnfv_or_sync_fifo_glo_rdfgras_rstslram_empty_i,
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD6_WRUU_WR_FIFOSUBCORE_FIF0_xsdm_v3_0_0_wrtffo_instInst_fifo_genlgconvffo_rdfgr_rfgnfv_or_sync_fifo_glo_wrigwas_wstslram_empty_i,
dbg_hubInstBSCANID_u_xsdm_idCORE_XSDB_UUT_MASTERU_ICON_INTERFACEU_CMD6_WRUU_WR_FIFOSUBCORE_FIF0_xsdm_v3_0_0_wrtffo_instInst_fifo_genlgconvffo_rdfgr_rfstbikrd_rst_reg[0],
dbg_hubInstBSCANID_u_xsdm_idSWITCH_N_EXT_BSCAN_bscan_instu_bscan_tms, u_ila_0Instila_core_instu_ila_regsU_XSDB_SLAVEis_daddr_q[14], u_ila_0Instila_core_instu_ila_regsU_XSDB_SLAVEis_daddr_q[14]
Select u_ila_0Instila_core_instu_ila_regsU_XSDB_SLAVEis_daddr_q[14] and the first 15 of 23 listed.